1 Lecture 12 OLS in Practice Short Answer Solutions

2 Question #1

Suppose that the one-way ticket price, $fare_i$, for airline ticket i is related to distance, measured in hundreds of miles, $dist_i$, by

$$\log(fare_i) = 4.666 + 0.043\, dist_i.$$

a. Write $fare_i$ as a function of $dist_i$ rather than $\log(fare_i)$.

Solution:

Exponentiating both sides,

$$fare_i = \exp\left(4.666 + 0.043\, dist_i\right).$$

b. What is the marginal effect of $dist_i$ on $fare_i$?

$$\frac{\partial fare_i}{\partial dist_i}$$

Solution:

$$\frac{\partial fare_i}{\partial dist_i} = \exp\left(4.666 + 0.043\, dist_i\right)(0.043) = 0.043\, fare_i.$$

c. Let

$$100\frac{\partial fare_i}{fare_i}$$

approximate the percentage change in $fare_i$. Show that the coefficient 0.043 implies that increasing distance by 1 mile, rather than increasing $dist_i$ by one unit (which corresponds to 100 miles), is associated with an approximate 0.043% increase in fare.

Solution:

See lecture notes.

3 Question #2

Suppose that the one-way ticket price, $fare_i$, for airline ticket i is related to distance, measured in hundreds of miles, $dist_i$, by

$$fare_i = 41 + 66\log(dist_i)$$

a. What is the marginal effect of $dist_i$ on $fare_i$?

$$\frac{\partial fare_i}{\partial dist_i}$$

Solution:

$$\frac{\partial fare_i}{\partial dist_i} = \frac{66}{dist_i}$$

b. Let

$$100\frac{\partial dist_i}{dist_i}$$

approximate the percentage change in $dist_i$. Re-write the marginal effect to indicate that the formula above shows that a 1% increase in $dist_i$ corresponds to a \$0.66 increase in $fare_i$.

Solution:

Re-arranging the marignal effect

$$\frac{\partial fare_i}{\frac{\partial dist_i}{dist_i}} = 66$$

$$\frac{\partial fare_i}{100\frac{\partial dist_i}{dist_i}} = \frac{66}{100} = 0.66$$

c. Show that the interpretation of the coefficient 66 does not depend on the units of $dist_i$. For example, if $dist_i$ were measured in miles instead of hundreds of miles, show that a 1% increase in distance would still increase $fare_i$ by approximately \$0.66.

Solution:

If $dist_i$ is measured in miles, then the original variable measured in hundreds of miles is $dist_i/100$. The model becomes

$$fare_i = 41 + 66\log(dist_i/100).$$

Using the properties of logarithms,

$$\begin{aligned} fare_i &= 41 + 66\left[\log(dist_i) - \log(100)\right] \\ &= (41 - 66\log(100)) + 66\log(dist_i). \end{aligned}$$

Changing the units of distance changes only the intercept. The slope coefficient on $\log(dist_i)$ remains 66, so its interpretation is unchanged: a 1% increase in distance is associated with an approximate \$0.66 increase in $fare_i$.

4 Question #3

Suppose that the one-way ticket price, $fare_i$, for airline ticket i is related to distance, measured in hundreds of miles, $dist_i$, by

$$\log(fare_i) = 4.25 + 0.40\log(dist_i).$$

a. Write $fare_i$ as a function of $dist_i$ rather than $\log(fare_i)$.

Solution:

See the lecture notes.

b. What is the marginal effect of $dist_i$ on $fare_i$?

$$\frac{\partial fare_i}{\partial dist_i}$$

Solution:

See the lecture notes.

c. Let

$$100\frac{\partial fare_i}{fare_i} \qquad \text{and} \qquad 100\frac{\partial dist_i}{dist_i}$$

approximate the percentage changes in $fare_i$ and $dist_i$, respectively. Show that the coefficient 0.40 implies that a 10% increase in distance is associated with an approximate 4% increase in fare.

Solution:

See the lecture notes.